

Helix Genomic Reference Laboratory

CAP-accredited / CLIA-certified reference laboratory
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PEDIATRIC CNS TUMOR PROFILE — NGS + RNA SIGNATURE (172-GENE) — FINAL REPORT

PATIENT WALKER, AIDEN T. MRN MRN-7720193 DOB 2017-11-18 8 y / Male	SPECIMEN Reference accession HX-NGS-CNS-018367 Client accession PED-26-N-00882 Frozen tumor + FFPE block B3 Collected 2026-03-12	REQUESTED BY Children's Medical Center of the Lakes N. Halpern, MD Received 2026-03-16 Reported 2026-03-26
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CLINICAL HISTORY (PROVIDED BY ORDERING SITE)

Previously well 8-year-old boy with three-week history of progressive morning headaches, ataxic gait, and one episode of emesis. MRI brain showed a 3.1 cm midline cerebellar vermis mass with obstructive hydrocephalus. No prior CNS history. No personal or family history consistent with Gorlin syndrome (no basal cell carcinomas, no rib or jaw anomalies, no palmar pits). Underwent posterior fossa craniotomy with subtotal resection on 2026-03-12. External ventricular drain placed peri-operatively.

RESULT SUMMARY

POSITIVE — **PTCH1** nonsense variant p.(Arg691*) detected (Tier I, classifying for SHH activation); **TP53** and **SUFU** wild-type by Sanger confirmation; RNA expression signature centroid: **MB, SHH-activated**; **MYC** and **MYCN** diploid (no amplification).

DETECTED SEQUENCE VARIANTS

Gene	Variant (HGVS)	Consequence	VAF	Tier
PTCH1	NM_000264.5:c.2071C>T p.(Arg691*)	Nonsense, premature stop; loss-of-function in SHH negative regulator	49%	I

Tier I = strong clinical / classifying significance. PTCH1 loss-of-function activates the SHH pathway and is one of the definitional events of SHH-activated medulloblastoma.

WILD-TYPE CONFIRMATIONS (PROGNOSTICALLY IMPORTANT)

Gene	Method	Result	Significance
TP53	NGS panel + Sanger confirmation (exons 4-10)	WILD-TYPE	In SHH-activated MB, TP53 status stratifies favorable (WT) vs adverse (mut) risk. This case is TP53 wild-type.
SUFU	NGS panel; orthogonal MLPA for large deletions	WILD-TYPE	Alternative SHH-activator excluded.

COPY-NUMBER ANALYSIS (FROM PANEL DATA)

Locus	Copy state	Interpretation
MYC (8q24.21)	Diploid (2 copies)	NO amplification. Excludes Group 3 high-risk MYC-amp subset.
MYCN (2p24.3)	Diploid	NO amplification.
GLI2 (2q14.2)	Diploid	NO amplification.
Chromosome 17	Diploid	No i(17q) (a Group 3/4 signature).
Chromosome 10	Diploid	No -10 (not relevant for MB classification).

RNA EXPRESSION-BASED SUBGROUP CENTROID

Centroid	Correlation	Call
MB, SHH-activated	0.91	MATCH
MB, WNT-activated	0.04	no
MB, Group 3	0.03	no
MB, Group 4	0.02	no

The expression-based call is consistent with the methylation classifier result (reported separately, accession BTMRS-26-M-1014).

INTERPRETATION

This pediatric CNS panel identifies a **PTCH1 p.(Arg691*) loss-of-function variant** (VAF 49%), a definitional event for SHH-activated medulloblastoma, together with a confirmatory **SHH-activated RNA expression signature**. **TP53** is wild-type (Sanger confirmed); **SUFU** is wild-type. There is no **MYC** or **MYCN** amplification. The constellation supports an integrated diagnosis of **medulloblastoma, SHH-activated and TP53-wildtype** per WHO CNS5 (2021). TP53 wild-type status stratifies this case into the favorable molecular risk category within SHH-activated medulloblastoma (the prognostically adverse SHH-activated and TP53-mutant entity is excluded).

Correlation with morphology, IHC pattern, and the methylation classifier result is recommended for the final integrated diagnosis.

METHODOLOGY & LIMITATIONS

Targeted NGS of 172 CNS-tumor-associated genes performed on tumor DNA extracted from frozen tissue and FFPE block (B3). Mean target coverage 1,020x; minimum 250x. Analytical sensitivity approximately 5% VAF for SNVs and small indels. RNA expression centroid analysis performed on RNA from the same specimen using a 285-gene MB-centroid panel. Copy-number analysis derived from per-gene normalized read depth. The assay does not reliably detect chromothripsis, complex rearrangements, or methylation events (reported separately via Illumina EPIC array). Helix Genomic Reference Laboratory validated this laboratory-developed test; it has not been cleared or approved by the FDA.

Electronically signed: Ranganathan, Vimal S., MD, PhD, Laboratory Director, and Park, R., PhD, Variant Scientist — 2026-03-26. Results released to ordering institution for integration into the combined diagnostic report.

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